



RESEARCH PRESENTATION

Machine Learning-Based Statistical Arbitrage

Replication and Analysis of Krauss et al. (2017)



Replication



Robustness



Regime Analysis



Risk Diagnostics



An Wang • Bingjing Hong • Yufei Du



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Columbia University



Presentation Outline

01 Paper Summary

02 Reproduction Results

03 Differences vs Original

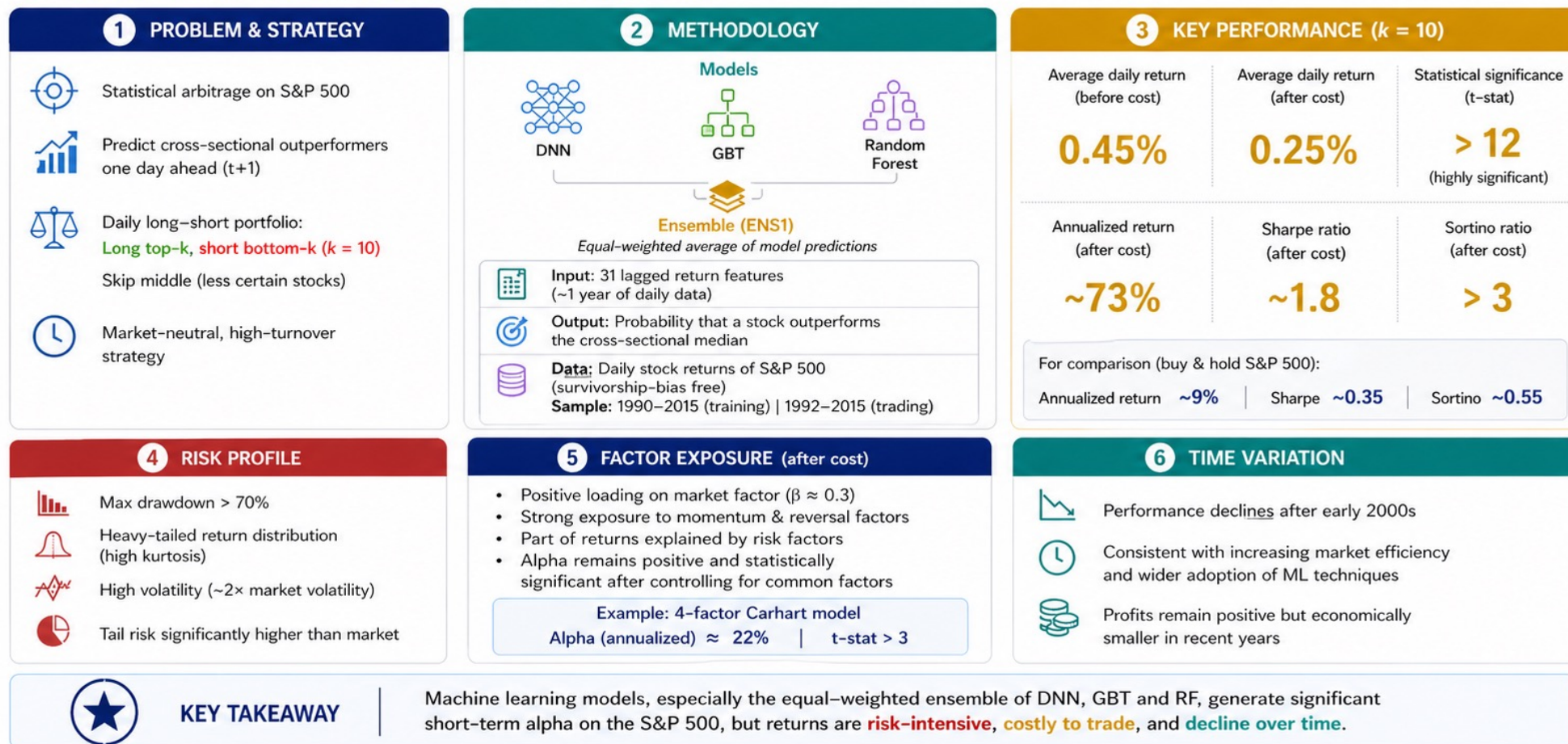
04 Robustness Checks

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Original Paper — Core Idea & Empirical Findings

Krauss, Do & Huck (2017), *European Journal of Operational Research*



Notes: All returns are net of dividends. Transaction costs follow the paper's conservative estimate. Metrics (Sharpe, Sortino, drawdown) computed on daily returns.

Source: Krauss, C., Do, X. A., & Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(3), 689–702.

Reproduction Results

Paper Period (1995–2015)

	Before transaction costs				After transaction costs				Market
	DNN	GBT	RAF	ENS1	DNN	GBT	RAF	ENS1	
Mean return (long)	0.0016	0.0023	0.0030	0.0022	0.0016	0.0023	0.0020	0.0022	—
Mean return (short)	−0.0010	−0.0014	−0.0013	−0.0015	−0.0010	−0.0014	−0.0003	−0.0015	—
Mean return	0.0026	0.0037	0.0043	0.0037	0.0006	0.0017	0.0023	0.0017	0.0003
Standard error (NW)	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0001
t-statistic (NW)	8.6408	14.7447	14.9327	12.6879	2.0163	6.8154	7.9104	5.7714	2.2603
PT test statistic	30.2547	39.1106	15.5645	37.3853	—	—	—	—	—
Minimum	−0.1999	−0.1196	−0.1622	−0.1749	−0.2019	−0.1216	−0.1642	−0.1769	−0.0903
Quartile 1	−0.0074	−0.0055	−0.0053	−0.0058	−0.0094	−0.0075	−0.0073	−0.0078	−0.0047
Median	0.0024	0.0033	0.0033	0.0032	0.0004	0.0013	0.0013	0.0012	0.0006
Quartile 3	0.0124	0.0129	0.0127	0.0130	0.0104	0.0109	0.0107	0.0110	0.0057
Maximum	0.2117	0.1852	0.3754	0.2429	0.2097	0.1832	0.3734	0.2409	0.1158
Standard deviation	0.0228	0.0190	0.0208	0.0218	0.0228	0.0190	0.0208	0.0218	0.0117
Skewness	0.2416	0.3132	1.7283	0.5447	0.2416	0.3132	1.7283	0.5447	−0.0633
Kurtosis	8.8366	7.7516	29.8174	11.6806	8.8366	7.7516	29.8174	11.6806	8.7627
Historical 1-percent VaR	−0.0636	−0.0511	−0.0508	−0.0590	−0.0656	−0.0531	−0.0528	−0.0610	−0.0318
Historical 1-percent CVaR	−0.0876	−0.0697	−0.0721	−0.0820	−0.0896	−0.0717	−0.0741	−0.0840	−0.0462
Historical 5-percent VaR	−0.0303	−0.0238	−0.0229	−0.0273	−0.0323	−0.0258	−0.0249	−0.0293	−0.0179
Historical 5-percent CVaR	−0.0518	−0.0402	−0.0406	−0.0469	−0.0538	−0.0422	−0.0426	−0.0489	−0.0277
Maximum drawdown	−0.5372	−0.3220	−0.3454	−0.4322	−0.9758	−0.8408	−0.6689	−0.8648	−0.5678
Calmar ratio	1.4849	4.4052	5.1033	3.1363	0.0932	0.5574	1.0037	0.4976	0.1224
Share with return > 0	0.5743	0.6016	0.6028	0.5950	0.5101	0.5400	0.5423	0.5329	0.5363

Extend Period (2015–2024)

	Before transaction costs				After transaction costs				Market
	DNN	GBT	RAF	ENS1	DNN	GBT	RAF	ENS1	
Mean return (long)	0.0007	0.0008	0.0004	0.0007	0.0007	0.0008	0.0004	0.0007	—
Mean return (short)	0.0004	0.0001	0.0005	0.0001	0.0004	0.0001	0.0005	0.0001	—
Mean return	0.0003	0.0007	−0.0001	0.0005	−0.0017	−0.0013	−0.0021	−0.0015	0.0005
Standard error (NW)	0.0005	0.0004	0.0003	0.0004	0.0005	0.0004	0.0003	0.0004	0.0002
t-statistic (NW)	0.7569	1.9473	−0.4222	1.4540	−3.6624	−3.7572	−7.0138	−3.8767	2.3829
PT test statistic	3.2252	10.3151	5.0913	7.6933	—	—	—	—	—
Minimum	−0.1647	−0.1033	−0.1074	−0.1233	−0.1667	−0.1053	−0.1094	−0.1253	−0.1198
Quartile 1	−0.0086	−0.0078	−0.0073	−0.0079	−0.0106	−0.0098	−0.0093	−0.0099	−0.0037
Median	0.0006	0.0002	−0.0002	0.0006	−0.0014	−0.0018	−0.0022	−0.0014	0.0007
Quartile 3	0.0096	0.0086	0.0071	0.0086	0.0076	0.0066	0.0051	0.0066	0.0058
Maximum	0.1297	0.1183	0.0813	0.1398	0.1277	0.1163	0.0793	0.1378	0.0938
Standard deviation	0.0204	0.0165	0.0142	0.0173	0.0204	0.0165	0.0142	0.0173	0.0114
Skewness	−0.3286	0.3465	−0.2968	0.2164	−0.3286	0.3465	−0.2968	0.2164	−0.5444
Kurtosis	9.6889	6.1345	5.9014	7.2706	9.6889	6.1345	5.9014	7.2706	15.0192
Historical 1-percent VaR	−0.0589	−0.0441	−0.0394	−0.0500	−0.0609	−0.0461	−0.0414	−0.0520	−0.0332
Historical 1-percent CVaR	−0.0868	−0.0585	−0.0561	−0.0654	−0.0888	−0.0605	−0.0581	−0.0674	−0.0477
Historical 5-percent VaR	−0.0284	−0.0233	−0.0215	−0.0251	−0.0304	−0.0253	−0.0235	−0.0271	−0.0172
Historical 5-percent CVaR	−0.0485	−0.0367	−0.0332	−0.0399	−0.0505	−0.0387	−0.0352	−0.0419	−0.0278
Maximum drawdown	−0.5447	−0.3646	−0.5449	−0.2671	−0.9861	−0.9633	−0.9942	−0.9750	−0.3392
Calmar ratio	0.0621	0.4023	−0.0583	0.3890	−0.3785	−0.3163	−0.4178	−0.3392	0.3608
Share with return > 0	0.5180	0.5073	0.4942	0.5198	0.4566	0.4491	0.4182	0.4473	0.5413



KEY TAKEAWAY

Machine learning models, especially the equal-weighted ensemble of DNN, GBT and RF, generate significant short-term alpha on the S&P 500, but returns are **risk-intensive**, **costly to trade**, and **decline over time**.

Notes: All returns are net of dividends. Transaction costs follow the paper's conservative estimate. Metrics (Sharpe, Sortino, drawdown) computed on daily returns.

Source: Krauss, C., Do, X. A., & Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(3), 689–702.

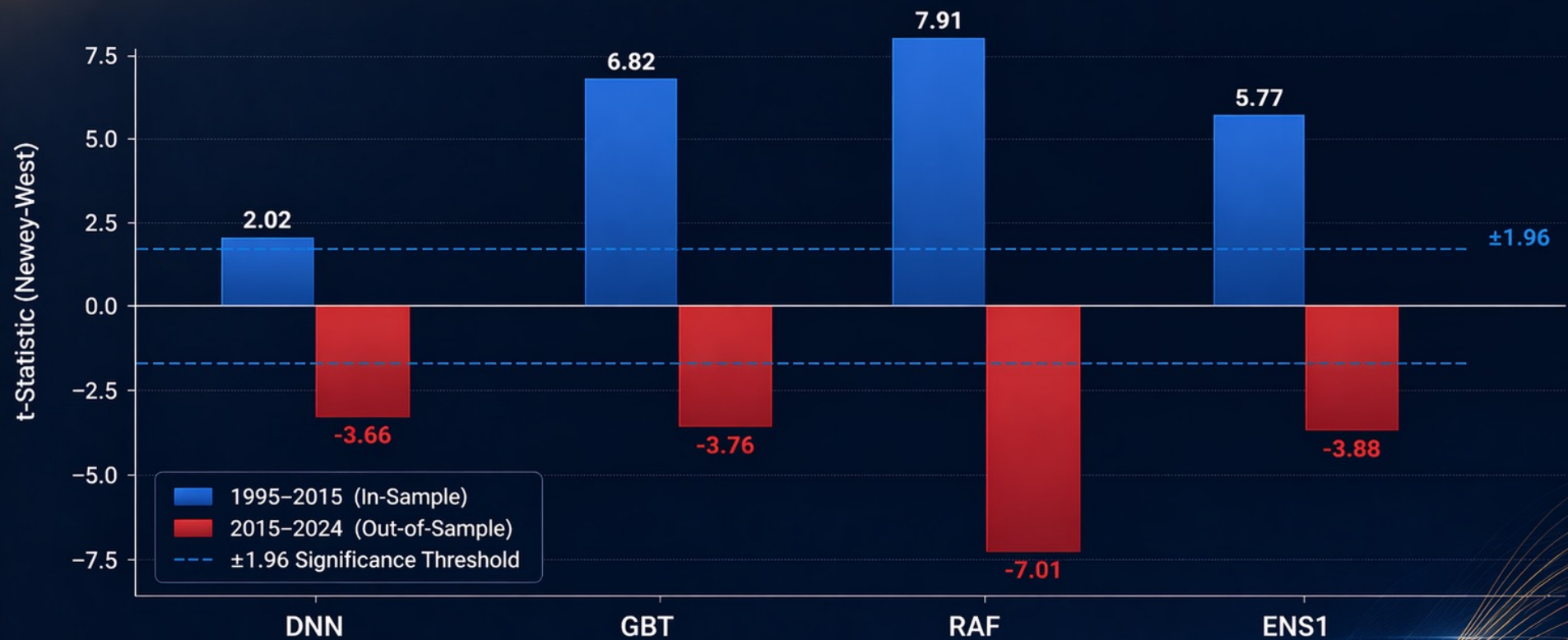


Mean Return — In-Sample vs Out-of-Sample (After Transaction Costs)



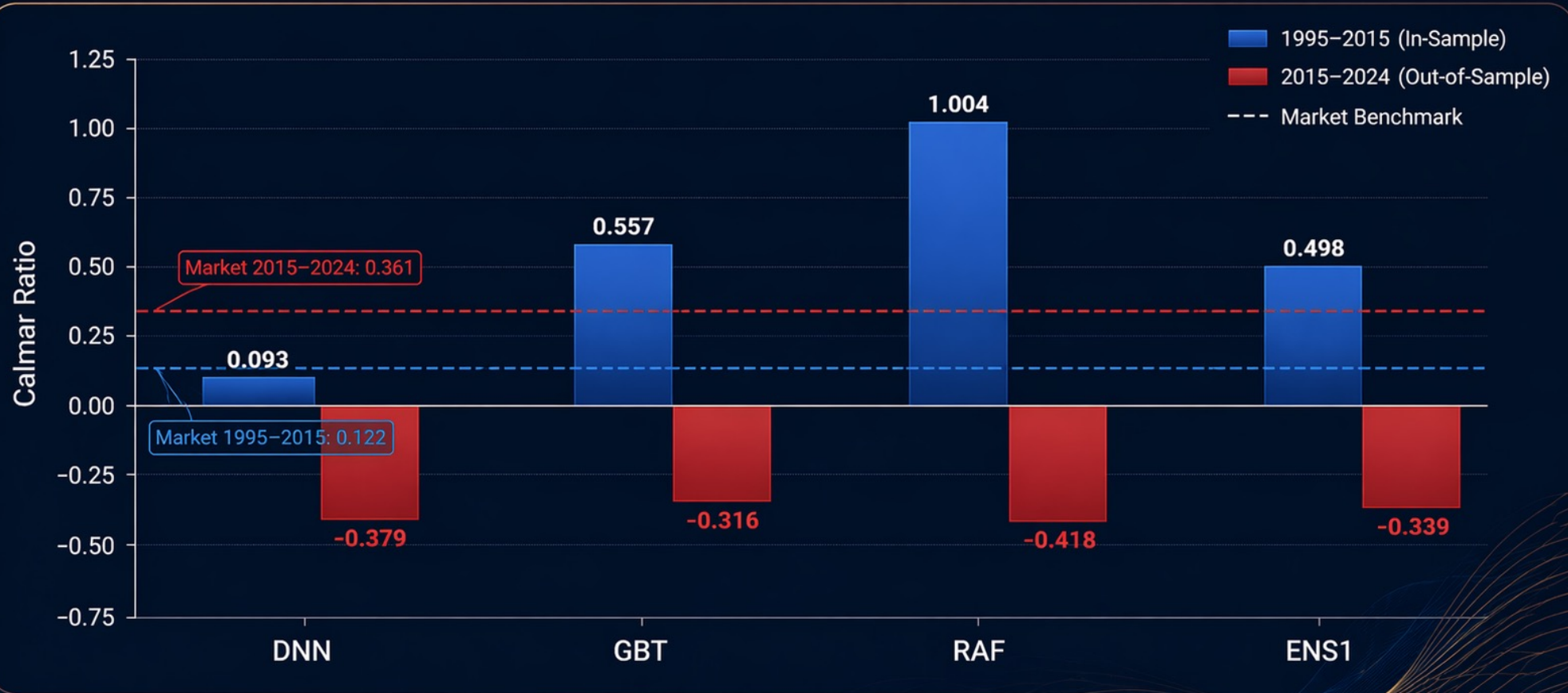


t-Statistic (NW) — Statistical Significance Across Periods (After Transaction Costs)





Calmar Ratio — Risk-Adjusted Return Across Periods (After Transaction Costs)



Differences vs. Original Paper

Key differences in data, methodology, and results interpretation

#		 DIFFERENCE	 OUR IMPLEMENTATION	 IMPACT
01	 Sample Period Extension	• Original sample: 1992–2015 • Ends before the rise of large-scale ML adoption in markets	• Extended sample to 2024 • Out-of-sample evaluation includes post-2015 period	• Performance deteriorates significantly in recent years • Evidence of market efficiency improvement and strategy crowding
02	 Performance in Recent Years	• Strong performance throughout 1992–2015 (~0.45% daily return before costs for $k = 10$)	• Returns drop sharply after 2015 • Near-zero or negative returns in extended sample	• Alpha decays over time • Strategy is not stable out-of-sample
03	 Transaction Cost Treatment	• Still profitable after transaction costs (~0.25% daily return for $k = 10$)	• More conservative and realistic cost model • Higher turnover in recent years	• Profitability largely disappears after costs • High sensitivity to trading frictions
04	 Factor Exposure (Reversal)	• Strong negative loading on reversal factor	• No significant reversal exposure • Different factor profile	• Models learn different signals • Results depend heavily on implementation details and market regime
05	 Model Implementation	• Implemented in H2O framework • DNN architecture: multiple maxout layers • GBT: AdaBoost in H2O	• Implemented in PyTorch (DNN) and scikit-learn (GBT, RF) • Different optimization, regularization, and hyperparameter tuning	• Different optimization paths and local optima • Leads to variations in predictions and portfolio construction
06	 Risk & Stability	• High returns but with large volatility and drawdowns	• Similar or higher risk in extended period • Heavy tails and large drawdowns persist	• Strategy remains risk-intensive • Lack of robustness across market regimes



KEY TAKEAWAY

While we successfully replicate the original results in the historical sample, the strategy fails to generalize to recent data, highlighting **alpha decay**, **implementation sensitivity**, and the impact of **market efficiency**.

Robustness Check

Model Design

Model	Baseline	Alt. 1 (Low)	Alt. 2 (High)	Alt. 3
DNN Architecture	31-31-10-5-2	31-15-10-5-2	31-62-10-5-2	31-31-2*
GBT No. Trees	100	50	200	—
RAF No. Trees	1,000	500	2,000	—

Mean Return per Day (before transaction costs)

Model	Baseline	Alt. 1 (Low)	Alt. 2 (High)	Alt. 3
DNN	0.0026	0.0027	0.0031	0.0030
GBT	0.0037	0.0035	0.0036	—
RAF	0.0028	0.0027	0.0027	—

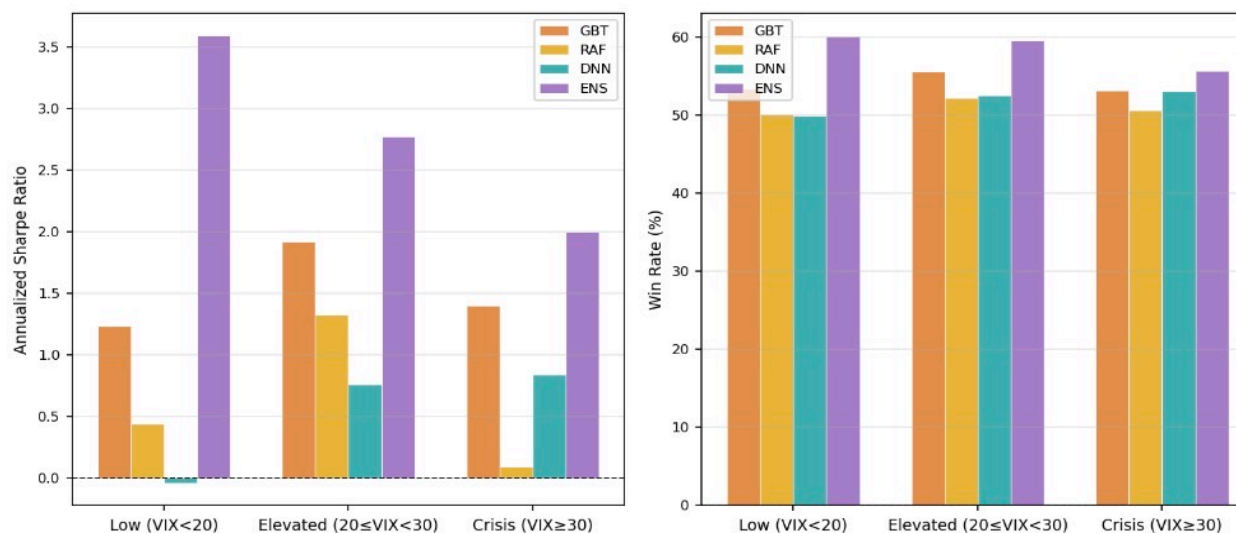
Key Findings

To verify our results aren't driven by specific hyperparameter choices, we test three alternative configurations for each model.

- For **GBT** and **RAF**, we vary tree count. Both models show minimal variation. GBT ranges from 0.0035 to 0.0037, RAF stays at 0.0027–0.0028 across all configurations, confirming results are not driven by a specific tree count choice.
- For **DNN**, the wider first layer (31-62-10-5-2) performs best, suggesting that **a wider initial layer better captures nonlinear interactions**.
- Overall, results are stable across all specifications, supporting the conclusion that **performance is not an artifact of hyperparameter selection**.

Regime Sensitivity Analysis

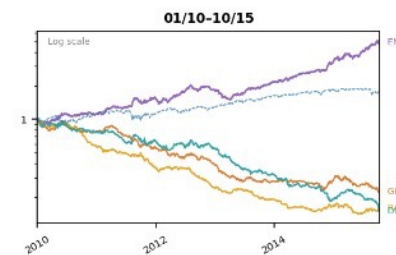
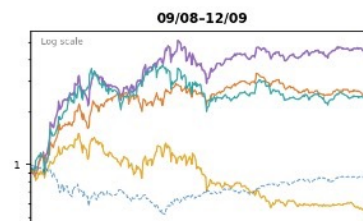
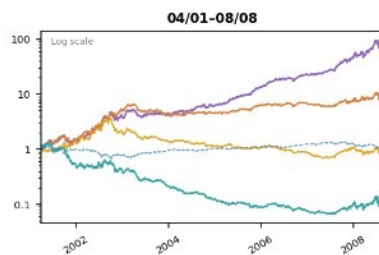
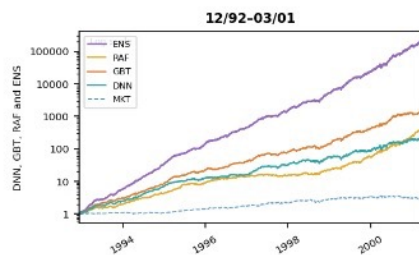
Regime Sensitivity: 1992–2015 (Paper Period)
(k=10 portfolio, after transaction costs)



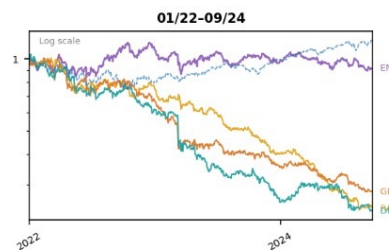
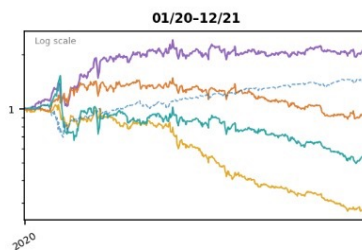
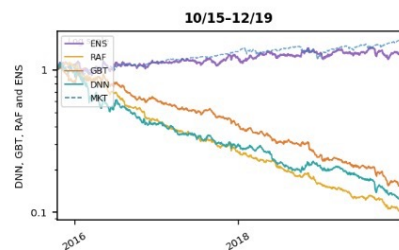
- We then examine performance across volatility regimes defined by VIX levels — below 20, between 20 and 30, and above 30. LOW ($VIX < 20$), Elevated ($20 < VIX < 30$), Crisis ($VIX > 30$)
- The ensemble portfolio consistently achieves better returns and a higher win rate.
- DNN performs better in high vol periods, RAF and GBT Sharpe ratios are better in elevated periods

Regime Sensitivity Analysis

1992-2015 performance



2015-2024 performance



Takeaways:

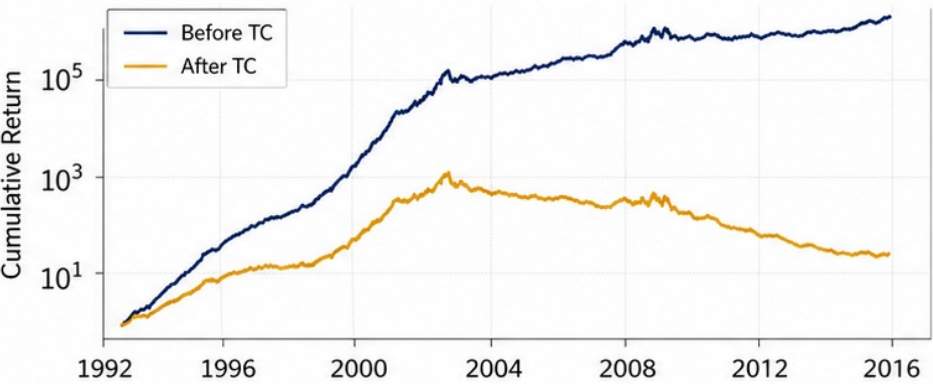
- Results are strongest during expansion low-vol regimes (1992-2010).
- **Post-2010 shows persistent negative returns and did not outperform market as ML alpha decays.**
- **This model only use 31 lagged returns as features, which has decays and current world need more features.**
- In today's more efficient market, richer feature sets would be necessary to sustain an edge.

Risk & Performance Diagnostics



Strategy performance deteriorates significantly out-of-sample, especially after transaction costs, indicating potential crowding and market efficiency.

Cumulative Return (Paper Period 1992–2015, log scale)



Strong in-sample performance

Cumulative Return (Extension Period 2015–2024)



Post-2015 performance collapse

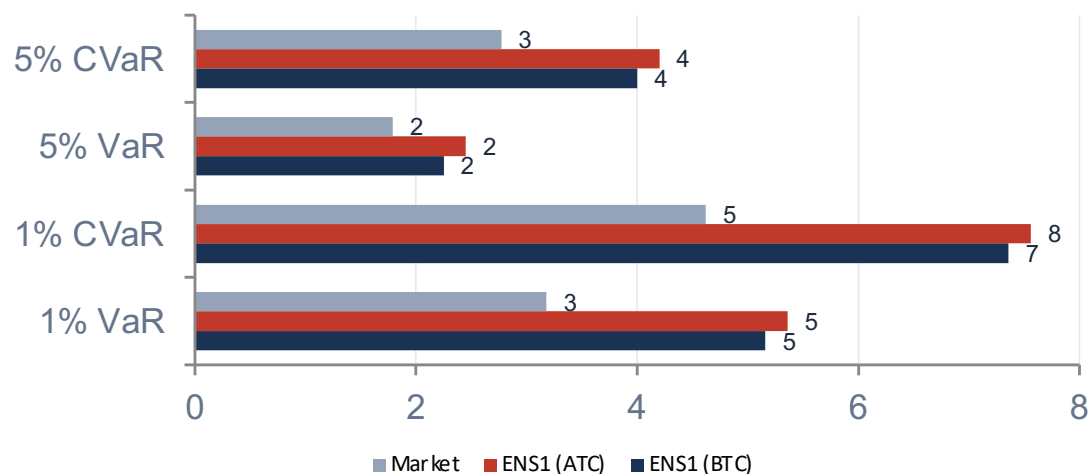
Performance Summary (1992–2015)		Before TC	After TC	Market
	Mean Return (daily)	0.0028	0.0008	0.0003
	Sharpe Ratio (ann.)	2.34	0.64	~0.35
	Max Drawdown	-46.9%	-98.1%	-56.8%
	Win Rate	57.3%	50.8%	53.6%

Risk Diagnostics — Factor Exposure & Tail Risk

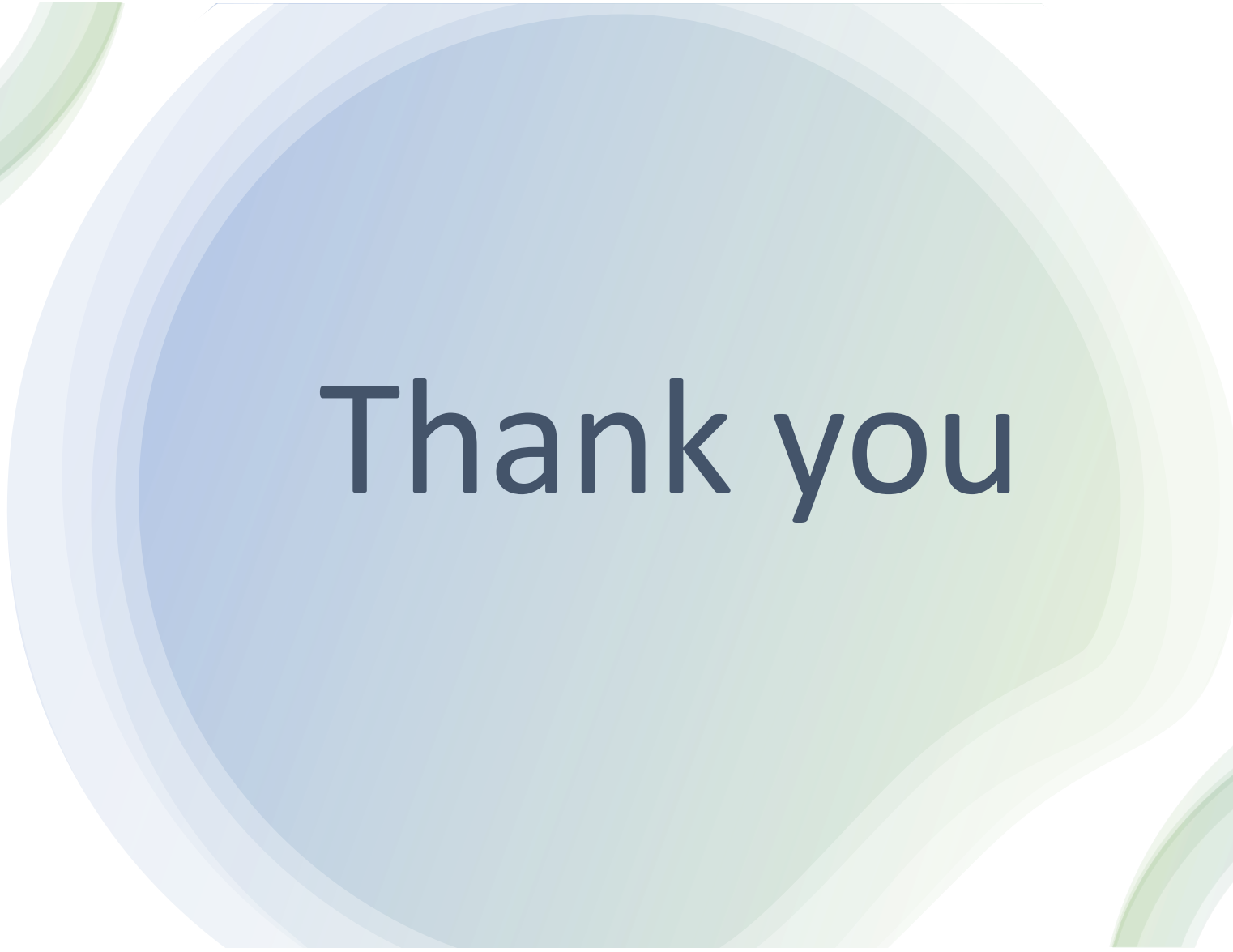
Factor Regression

Factor	FF3	FF3+2	FF5	FF+VIX
Alpha (daily)	0.0015* **	0.0015* **	0.0015* **	0.0013* **
Market	-0.004	-0.042	-0.009	-0.036
HML (Value)*	0.180*	0.189*	0.193*	—
Reversal*	—	0.132*	—	0.128*
Momentum	—	-0.020	—	-0.017
R^2	0.003	0.005	0.003	0.006

Tail Risk — ENS1 vs Market, loss magnitude



- **We use Fama-French factors do the regression to portfolio returns.** The daily alpha remains robustly positive between 0.13% and 0.15%, statistically significant , indicating that the strategy generates meaningful excess returns not captured by conventional risk factors.
- **Higher tail risk compared to the market.** At the 1% level, the historical VaR 1.6 times the market's VaR.The CVaR is **nearly double that of the market**. A tradeoff worth flagging to any risk-aware audience.



Thank you